

Nick DiSanto

<https://nickdisanto.github.io>

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Education	Vanderbilt University PhD in Computer Science GPA: 3.96/4.0. Advised by Dr. Ipek Oguz.	Jan. 2025 – Present
	California Baptist University B.S. in Computer Science, <i>summa cum laude</i> GPA: 4.0/4.0. Ranked #1 in graduating class.	Aug. 2019 – Apr. 2023
Research Experience	Vanderbilt Medical Image Computing Lab (MedICL) <i>Graduate Student Researcher</i> - Developing deep learning methods for image restoration and generation, with emphasis on 3D medical image volumes - Designing registration and volumetric inpainting models for high-resolution 3D OCT/OCTA data to correct motion-induced slice corruption - Exploring how learned representations can maintain robustness across diverse imaging tasks	Jan. 2025 – Present
	CalBaptist ML & NLP Lab <i>Lead Undergraduate Researcher</i> - Developed representation learning models to identify patterns in large-scale online data - Proposed a novel data preprocessing approach to help generalize medical imaging models - Led end-to-end research efforts across three papers, from inception through execution	Jan. 2022 – Dec. 2023
Industry Experience	GE Vernova <i>Software Engineer</i> - Led a wildfire forecasting initiative, including the end-to-end model training, visualization, and productization of ML pipelines for power-grid applications - Presented results of wildfire risk forecasting analytic to the C-Suite, contributing to secured funding for continued integration - Recognized as a 2024 GE Vernova Changemaker for co-leading a company-wide initiative to integrate GenAI into Electrification Software products	Dec. 2023 – Jan. 2025
	General Electric <i>Software Engineering Intern</i> - Led a database optimization initiative resulting in annual cost savings of several thousand dollars - Developed and optimized microservices supporting large-scale data ingestion pipelines	Jun – Aug 2022; Jun – Aug 2023
	Keysight Technologies <i>Software Engineering Intern</i> - Developed a web platform enabling clients to customize product configurations - Performed data analysis that eliminated 40% of redundant product-option configurations	Jun. 2021 – Aug. 2021

Publications	VAMOS-OCTA: Vessel-Aware Multi-Axis Orthogonal Supervision for Inpainting Motion-Corrupted OCT Angiography Volumes N DiSanto, EK Aghdam, H Liu, JJ Watson, YK Tao, H Li, I Oguz. SPIE Medical Imaging, 2026 [Oral Deep-Dive] [Best Student Paper]. [pdf] [code]		
	Monocular Absolute Depth Estimation from Endoscopy via Domain-Invariant Feature Learning and Latent Consistency H Li, D Lu, J d’Almeida, D Isik, EK Aghdam, N DiSanto, A Acar, S Sharma, JY Wu, RJ Webster III, Ipek Oguz. SPIE Medical Imaging, 2026 [Oral]. [pdf] [code]		
	From Geometry to Intensity: A Coarse-to-Fine Pipeline for Unsupervised 3D Ultrasound Stitching X Yao, R Yu, D Lu, N DiSanto, EK Aghdam, KJ Oguine, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz. SPIE Medical Imaging, 2026.		
	A Benchmark Study of Methods for Surgical Instrument Segmentation in Central Airway Endoscopy D Isik, KJ Oguine, N DiSanto, J d’Almeida, RJ Webster III, I Oguz, H Li. SPIE Medical Imaging, 2026 [Best Poster: Honorable Mention].		
	LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching X Yao, R Yu, N DiSanto, E Aghdam, KJ Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz. Medical Imaging with Deep Learning (MIDL), 2025 [Oral]. [pdf] [code]		
	Spatial Analysis of Social Media’s Proxies for Human Emotion and Cognition A Corso, N DiSanto, N Corso, E Lee. International Conference on Information, 2024.		
Teaching Experience	EGR121: Intro to C++, Teaching Assistant	SP 2023	
	EGR225: Discrete Structures, Teaching Assistant	FA 2022	
	CSC312: Algorithms, Tutor	SP/FA 2022	
	EGR329: Computer Architecture, Tutor	SP/FA 2022	
	PHY201: Physics for Engineers, Teaching Assistant	SP/FA 2021	
Awards & Honors	SPIE Medical Imaging Best Student Paper Award		Feb. 2026
	2024 GE Vernova Changemaker (selected out of 1200 nominees)		Nov. 2024
	CS Outstanding Student Award (ranked #1/40)		Apr. 2023
	Inducted into Alpha Chi Honor Society (top 10%)		Apr. 2022
	President’s List (every semester of undergraduate study)		FA 2019 – SP 2023
Professional Memberships	Institute of Electrical and Electronics Engineers (IEEE)		Jan. 2025 – Present
	Society of Photo-Optical Instrumentation Engineers (SPIE)		Jul. 2025 – Present